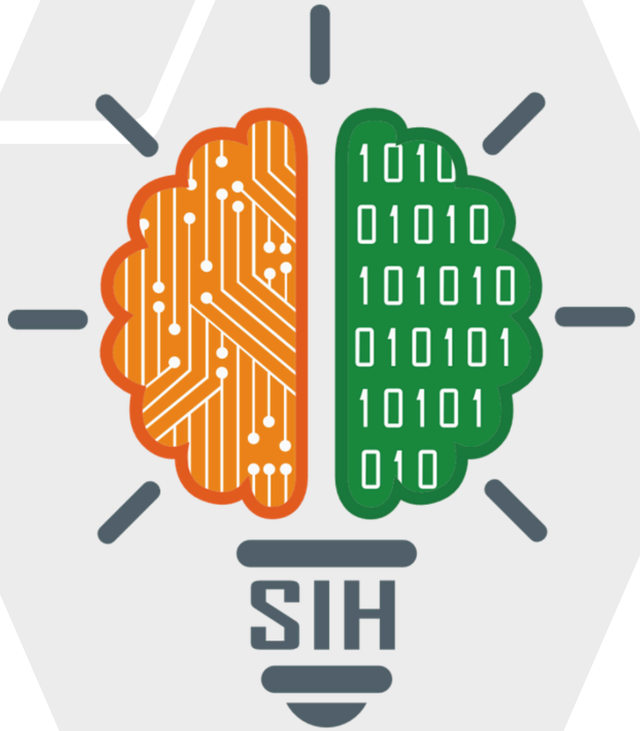
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TITLE PAGE

- **Problem Statement ID –** SIH26095
- **Problem Statement Title-** Smart Real-Time Monitoring & Inspection Mobile App
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team Name -** Technical Terminators



SATYA-NET

SCHEME-AWARE AI INTEGRITY & ADAPTIVE INSPECTION NETWORK

Every signal becomes evidence. Every anomaly becomes a verification task. Every finding remains open until reality is re-verified.

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PROPOSED SOLUTION

- **Real-Time Scheme Surveillance** – Live project status, CCTV feeds & monitoring dashboard.
- **AI-Based Surprise Inspection Allocation** – Risk-aware random assignment of PMU inspection duties.
- **Geo-Tagged Field Inspection** – GPS-verified snapshots, videos, checklists & inspection reports.
- **Random VC Beneficiary Verification** – Unannounced VC with Project Incharge, staff & beneficiaries.
- **AI Attendance & Anomaly Analytics** – Proxy attendance, reporting inconsistencies & operational anomaly detection.
- **Snapshot Cross Validation** – Time-stamped snapshots cross-validated against CCTV, attendance & reports with hashed evidence integrity.

SCHEME CONTROL FABRIC



Scheme-Specific Verification Logic

GOVERNANCE FLOW

- **INTENT** – Guidelines & KPIs
- **CONTROL** – Digital rules & checklists
- **PROOF** – Geo-tagged hashed submissions
- **INSIGHT** – Compliance & impact intelligence
- **ACTION** – Corrective action & re-verification
- **TRUST** – Accountability & trust recovery

NOVEL FEATURES

Tamper-Proof Cryptographic Digital Tagging

Cryptographically locks evidence artifact carrying Who + Where + What + When + Where + Session + Device + Integrity Hash

Cross-Source Validation Engine

Compares different data sources against each other to catch hidden contradictions

Dynamic AI Inspection Scheduler

Employs controlled randomization auto-assigning surprise visits by combining instituional risk score and inspection rotation history.

Uncertainty-Aware Verification Engine

Flags missing, old, or unclear evidence instead of instantly branding an institution as fraudulent

Beneficiary Reality Index (BRI)

Replaces generic feedback with a quantifiable ground-truth metric

Institutional Trust Twin

Builds a time-evolving Trust Trajectory computing institutional reliability

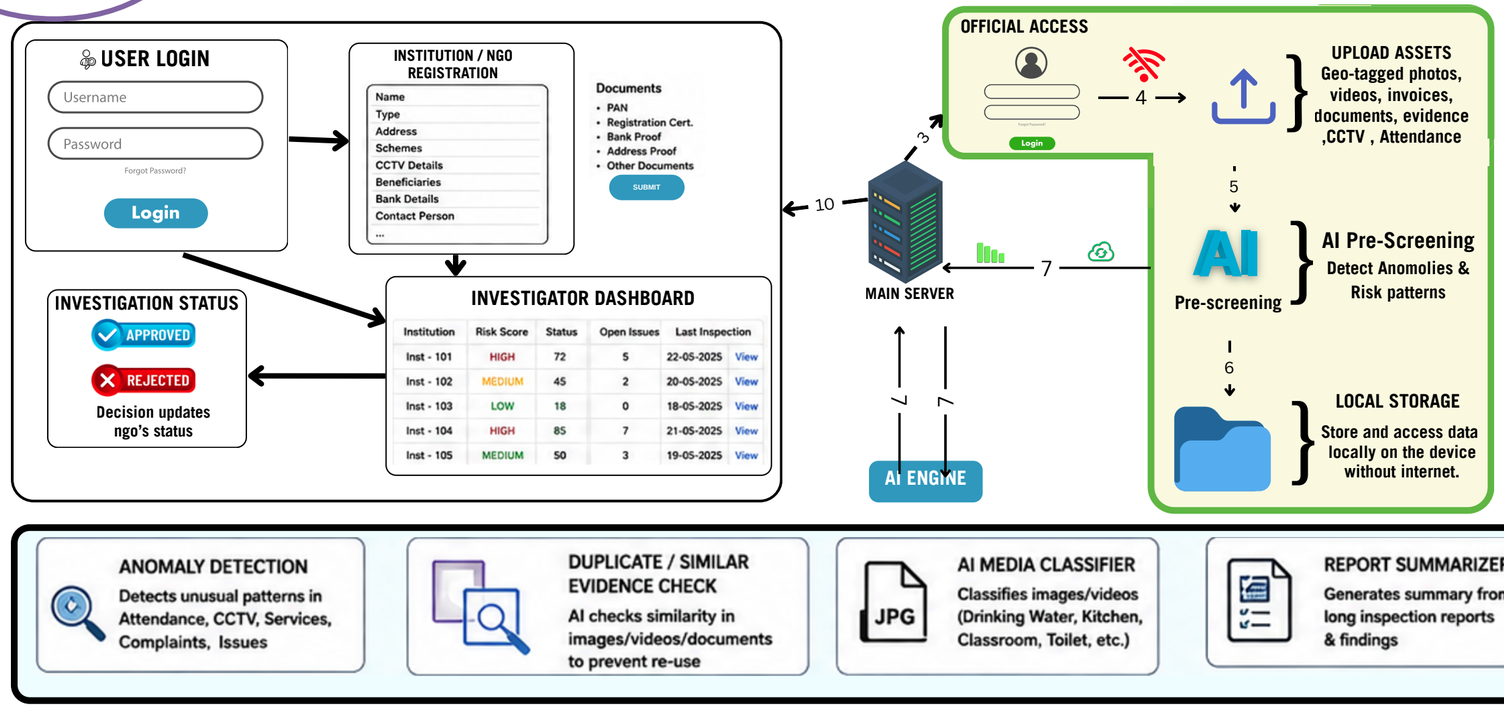
OVERCOMING EXISTING SYSTEM LOOPHOLES

- Fragmented & Report-Centric Monitoring
- Generic & Single-Source Verification
- Predictable & Biased Inspections
- Static Risk & Unverified Closure
- Institution-Centric & One-Time Monitoring

Technical Terminators

TECHNICAL OVERVIEW

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TECH STACK

AI / ML	• Python • Scikit-learn • OpenCV • PyTorch
Frontend	• Flutter • Dart • Glassmorphism
Backend	• Node.js • Express.js • Socket.IO
DataBase	• PostgreSQL • PostGIS • Redis
Infra	• AWS - EC2 • Docker

ANOMALY DETECTION
Detects unusual patterns in Attendance, CCTV, Services, Complaints, Issues

DUPLICATE / SIMILAR EVIDENCE CHECK
AI checks similarity in images/videos/documents to prevent re-use

AI MEDIA CLASSIFIER
Classifies images/videos (Drinking Water, Kitchen, Classroom, Toilet, etc.)

REPORT SUMMARIZER
Generates summary from long inspection reports & findings

01 SCHEME CONTROL FABRIC
Scheme Rules → Verifiable Controls + Evidence Logic

02 MULTI-MODAL EVIDENCE INGESTION
CCTV + VC + GPS + Attendance + Media + Documents → Evidence Stream

03 EVIDENCE INTEGRITY ENVELOPE
Evidence → Source + Time + Location + Device + Session Integrity

04 EVIDENCE RELATIONSHIP GRAPH
Entities + Evidence → Concordance / Discordance Map

05 AI INTEGRITY ENGINE
Evidence Graph → Anomalies + Uncertainty + Integrity Priority

06 VERIFICATION ORCHESTRATION ENGINE
Risk Signal → WHO + WHEN + WHERE + HOW to Verify

07 FIELD INTELLIGENCE APPLICATION
Mission → Geo-Verified Inspection + Live Ground Evidence

08 CORRECTIVE ACTION & REVERIFICATION
Finding → Action → Evidence → Re-verification → Closure

09 INSTITUTIONAL TRUST TWIN
History + Outcomes → Dynamic Institutional Trust Trajectory

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FEASIBILITY AND VIABILITY

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TECHNICAL FEASIBILITY: GPS and camera-based AI are all proven, ready-to-use tech, nothing experimental needed.

TECHNICAL VIABILITY: Built in modules, so parts like CCTV or scoring can be upgraded without touching the rest.

OPERATIONAL FEASIBILITY: Same inspection process as today, just on a phone checklist and photo instead of paper.

OPERATIONAL VIABILITY Auto-reminders cut down follow-up work and keep records clean.

ECONOMIC FEASIBILITY: Runs on open-source tools and is cheap to build and pilot.

ECONOMIC VIABILITY Catches fraud and cuts wasted visits and pays for itself within a year.

FEASIBILITY VS. VIABILITY

CHALLENGES

- Rural CCTV/network gaps
- GPS unreliable indoors
- Inspector–institution collusion
- Beneficiaries without smartphones
- False alarms on genuine institutions

MITIGATIONS

- Offline app and event-triggered clips
- QR/NFC as backup presence proof
- Independent beneficiary SMS checks
- Low-tech IVR calls, no app needed
- Explainable flags & appeal process

Implementation Strategy

- **Phase 1 (Pilot):** Roll out in 1 district under DoSJE, covering 20–30 institutions.
- **Phase 2 (Optimization):** Add CCTV cross-checks, video-call verification, and fraud-risk scoring.
- **Phase 3 (Nationwide Rollout):** Extend to all institutions, link with fund-disbursement data for full ghost-project detection.

Goal: 6–9 months full deployment cycle.

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IMPACT AND BENEFITS

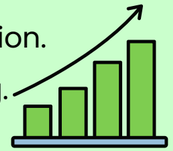
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Increase :

Evidence Confidence — +40% stronger evidence validation.

Inspection Precision — +35% better inspection targeting.

Real-Time Visibility — +50% faster issue visibility.



Decrease :

40% fewer unresolved reporting mismatches.

35% reduction in attendance/service inconsistencies.

60% manual case screening effort.

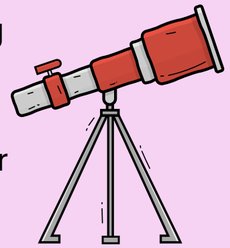


Future Scope :

Automate - Introduce predictive risk forecasting, automated verification planning and advanced CCTV intelligence to proactively identify emerging issues.

Expand - Extend beneficiary verification with multilingual IVR, advanced Trust Twins and deeper scheme-specific intelligence.

Connect - Integrate additional DoSJE systems, approved data sources, CCTV networks and cross-scheme evidence for unified monitoring.



Impact

Time to Truth Acceleration — Turns delayed reports into early, actionable ground-reality alerts.

Rupee to Outcome Integrity — Verifies each funded claim against actual beneficiary-level service delivery.

Compliance Lag Elimination — Collapses discovery-to-correction cycles before failures become systemic.

Leakage Before Loss Prevention — Intercepts unsupported utilisation before money becomes unverified compliance.

Last Mile Reality Guarantee — Makes beneficiary experience the final proof of delivery.

Benefits of the Solution

Governance - Defensible, explainable verification decisions.

Social - Beneficiary experience becomes measurable evidence.

Operational - Inspection capacity focused where verification matters.

Accountability — Issues remain open until independently verified.

Institutional — Compliance improvement can restore trust.

Technical Terminators

RESEARCH AND REFERENCES

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1. Sultani, W., Chen, C., & Shah, M. (2018). Real-world anomaly detection in surveillance videos. In 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 6479–6488), vol. 200, art. 103313, 2022. [DOI-link]

2. Yan, C., Li, B., Vorobeychik, Y., Laszka, A., Fabbri, D., & Malin, B. (2018). Get your workload in order: Game theoretic prioritization of database auditing, vol. 198, art. 116889, 2022 [DOI-link]

3. Kounas, D., Voutyras, O., Palaiokrassas, G., Litke, A., & Varvarigou, T. (2020). QuietPlace: An ultrasound-based proof of location protocol with strong identities. Applied System Innovation, 3(2), 19, vol. 557, [DOI-link]

4. Ribnick, E., Atev, S., Masoud, O., Papanikolopoulos, N., & Voyles, R. (2006). Real-time detection of camera tampering in video surveillance. vol. 45, no. 7, art. 102186, 2021. [DOI-link]

5. Kandasamy, N., Vorobeychik, Y., & Laszka, A. (2026). Incentive-aware AI safety via strategic resource allocation: A Stackelberg security games perspective., vol. 118, art. 108509, 2022. [DOI-link]

6. Saroiu, S., & Wolman, A. (2018). Physical location verification and trusted sensor data processing in mobile devices. U.S"Detecting fake and manipulated multimedia content using AI techniques," Multimed. Tools Appl., vol. 82, pp. 13725–13744, 2023. [DOI-link]